Data-Driven Clothing for Interactive Applications

Machine Learning Mini-Project

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Tools used: Python, Google Colab, NumPy, Pandas, Scikit-learn, PyTorch and Matplotlib

1. Introduction

Clothing simulation is used in applications such as games, animation and virtual environments. Traditional physics-based simulation can require a lot of computation. In this mini-project, we implemented a simplified machine-learning approach inspired by Kangrui Xue and Jane Wu's paper, *Data-Driven Clothing for Interactive Applications* (Stanford CS229 Student Project, Spring 2021). The aim was to train a model to predict clothing-deformation values from human-pose parameters.

2. Dataset

The original numerical dataset from the reference paper was not available for our implementation, so we generated 2,000 synthetic, simulation-inspired samples. Each pose contains 56 input features, representing 14 joints with four quaternion values per joint. The clothing output contains 6,510 values, representing 2,170 vertices with three coordinates per vertex. We divided the samples into 1,200 training samples, 400 validation samples and 400 test samples. These synthetic samples are useful for demonstrating the workflow, but they are not a physically accurate cloth simulation.

3. Methodology

First, we prepared the pose inputs and clothing-deformation outputs. Since the output has 6,510 values, we used Principal Component Analysis (PCA) to reduce its dimensionality. PCA was fitted on the training outputs and reduced them to 128 components. In this implementation, these components retained approximately 99.9829% of the variance. The models predicted the 128 PCA values, and inverse PCA was used to reconstruct the 6,510 output values.

3.1 Models and training

Linear Regression was used as a baseline model. We also built a fully connected neural network in PyTorch. It has 56 input units, hidden layers of 128 and 256 units with ReLU activation, and an output layer of 128 units. The network was trained with Mean Squared Error (MSE) loss and the Adam optimizer. The learning rate was 0.003, the batch size was 32, and training ran for 100 epochs. We compared the models using test-set MSE in PCA space.

Table 1. Summary of the dataset and representation

Property	**Value**
Generated samples	2,000
Pose input features	56 (14 joints × 4 quaternion values)
Clothing output features	6,510 (2,170 vertices × 3 values)
Train / validation / test	1,200 / 400 / 400
PCA components / variance retained	128 / approximately 99.9829%

4. Results and discussion

The neural network achieved a lower test MSE than Linear Regression on our synthetic dataset. Its PCA-space test MSE was 0.0082453, compared with 0.0095094 for Linear Regression, which is a reduction of about 13.29%. The neural network's test MSE after reconstruction to the original output space was 0.0001621. These values describe our implementation and should not be interpreted as the numerical results of the original paper.

5. Implementation and demonstration

The work was implemented in a Google Colab notebook using Python and the listed machine-learning libraries. The notebook includes the data preparation, PCA transformation, model training, evaluation and a test-sample prediction demonstration. The saved model and results table are included with the project files.

6. Limitations

The main limitation is that the original dataset was unavailable, so synthetic data was used. The generated deformations do not reproduce the physical behaviour of real cloth. The dataset is also smaller and the implementation is simpler than a full cloth-simulation system. Therefore, the project demonstrates the machine-learning workflow rather than reproducing the original paper's full system or results.

7. Conclusion

This project gave us a practical way to explore how machine learning can map human-pose inputs to clothing-deformation outputs. PCA reduced the output space from 6,510 values to 128 components, and the neural network performed better than the Linear Regression baseline on the synthetic test data. Future work could use a real cloth-simulation dataset and evaluate the model on more varied poses.

Table 2. Model evaluation results

Evaluation metric	**Result**
Linear Regression test MSE (PCA space)	0.0095094
Neural Network test MSE (PCA space)	0.0082453
MSE reduction compared with baseline	13.29%
Neural Network test MSE (original space)	0.0001621

Reference: Kangrui Xue and Jane Wu, “Data-Driven Clothing for Interactive Applications,” Stanford CS229 Student Project, Spring 2021.